ALP BERKE ARDIC

Chicago, IL | alpberke.ardic@gmail.com | (312) 889-4184 | linkedin.com/in/alpberkeardic

SUMMARY

Ph.D. candidate in ECE at the University of Illinois Chicago with 3+ years in distributed and decentralized ML, federated learning, and communication-efficient deep learning at the edge. Published at IEEE MILCOM and LANMAN, with two papers under IEEE Transactions review. Currently working on Wildfire Hotspot Detection Project and Distributed World Models. Seeking an ML research internship.

EDUCATION

Ph.D. in Electrical and Computer Engineering, University of Illinois Chicago *Aug 2022 - Present*
GPA: 4.0/4.0 | Focus: Distributed & Decentralized ML, Federated Learning | Advisor: Prof. Erdem Koyuncu
Teaching Assistant for Machine Learning, Deep Learning, and Pattern Recognition; mentored 100+ students.

M.Sc. in Electrical and Computer Engineering, University of Illinois Chicago *Aug 2022 - Apr 2026*
GPA: 4.0/4.0

B.Sc. in Electrical and Electronics Engineering, Bilkent University, Ankara, Turkey *2017 - 2022*

RESEARCH & PROFESSIONAL EXPERIENCE

Graduate Research Assistant, University of Illinois Chicago, Chicago, IL *Sept 2022 - Present*

- Developed Random Walking Snakes (RWS), a distributed training framework with 44% faster convergence than Random Walk Learning; extended to LLM fine-tuning on non-IID data for 30% time gains and robust accuracy convergence.
- Designed the synchronous and asynchronous Decentralized Model-Distributed Inference algorithms that partition ML models across heterogeneous edge devices via conflict-free graph coloring approach, cutting communication latency >40%.
- Validated on 5G edge testbeds, cutting bandwidth use 25% while preserving model accuracy across multi-node deployments.

Research Intern, TUBITAK SAGE, Ankara, Turkey *Sept 2021 - May 2022*

- Developed real-time object detection algorithms (YOLO-based) for fast-moving targets, improving detection accuracy by 18% under motion blur via custom data augmentation and hyperparameter tuning.
- Optimized the model inference preprocessing pipeline for a >25% real-time speedup, improving system efficiency and reliability in high-speed imaging environments.

Embedded & Camera Systems Intern, ASELSAN, Ankara, Turkey *Jun 2021 - Aug 2021*

- Built a complete embedded camera interface system from the ground up on a microcontroller in C, handling image capture, processing, and display.
- Designed a custom GUI and integrated on-board memory management to save photos directly on the device.

Robotics Researcher, Bilkent Miniature Robotics Lab, Ankara, Turkey *Jan 2021 - Nov 2021*

- Developed low-cost UAV systems for infrastructure inspection; contributed to the SCoReR collision-resilient drone project (acknowledged; IEEE Xplore https://ieeexplore.ieee.org/document/10121952).

AI & Computer Vision Intern, TWIN Science & Robotics, Istanbul, Turkey *Jul 2019 - Sept 2019*

- Built real-time object-tracking vision pipelines on Raspberry Pi and Arduino for TWIN Science kits, part of the "Science Migration to Anatolia" project bringing hands-on science to children in rural areas.

PUBLICATIONS

- [1] A. B. Ardic, M. Dwyer, E. Koyuncu, “Decentralized Model-Distributed Inference,” 2025 IEEE Military Communications Conf. (MILCOM), Los Angeles, CA. https://ieeexplore.ieee.org/document/11310356
- [2] A. B. Ardic, H. Seferoglu, S. E. Rouayheb, E. Koyuncu, “Random Walking Snakes for Decentralized Learning at Edge Networks,” 2023 IEEE LANMAN Symposium, London, UK. https://ieeexplore.ieee.org/document/10189426
- [3] A. B. Ardic, H. Seferoglu, S. El Rouayheb, E. Koyuncu, “Random Walking Snakes for Decentralized Learning of Heterogeneous Data at Edge Networks,” under review at IEEE Transactions on Mobile Computing (TMC).
- [4] A. B. Ardic, M. Dwyer, E. Koyuncu, “Cost-Aware Asynchronous Persistent Graph Coloring for Decentralized Model-Distributed Inference,” submitted to IEEE TPDS.

SELECTED PROJECTS

On-Device CNN Compression on ESP32 Microcontroller

- Compressed a CNN onto an ESP32 (312KB DRAM) using structured pruning, adaptive pooling, and 8-bit quantization for on-device inference under tight edge constraints.

TECHNICAL SKILLS

ML & Deep Learning: Distributed & Decentralized ML, Federated Learning, Model Compression, LLM Fine-Tuning, Transformers, Computer Vision, GenAI, Reinforcement Learning
Systems & Deployment: CUDA/cuDNN, Edge Deployment, Comm-Efficient Learning, Parallel Computing, AWS
Frameworks & Languages: PyTorch, TensorFlow, Hugging Face, OpenCV | Python, C/C++, MATLAB, Linux, Arduino